

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



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Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

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Abstract

Falls, frailty, and chronic diseases such as cardiovascular disease, diabetes, and obesity are major contributors to morbidity and mortality among older adults. Existing predictive models often rely on limited physical health variables, with under-representation of nutrition and lifestyle factors that may enhance early risk detection. This study develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in adults aged ~ 50 and above. The approach integrates physical, mental, lifestyle, and socio-demographic predictors, with attention to under-explored factors identified in prior literature. Multiple modelling strategies, including logistic regression, random forests, and gradient boosting, will be compared using rigorous cross-validation and feature importance analysis. By identifying high-impact modifiable predictors, this research aims to improve early risk stratification, support targeted prevention strategies, and inform policy for healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this paper without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This paper has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor and Dr. Name Surname at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well-organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

DAPI 4',6-diamidino-2-phenylindole; a fluorescent stain that binds strongly to DNA and serves to mark the nucleus in fluorescence microscopy

IMU Inertial Measurement Unit

TILDA The Irish Longitudinal Study on Ageing

TRIAD ejfowejfopjewopfjweopfjopwejfopwjeopfjweopfjopwefjopwej

WIMU Wireless Inertial Measurement Unit

LIST OF ABBREVIATIONS

1

Introduction

TILDA Here you can write you own text . . . also we can write a footnote like this.¹

This document offers few basic examples on how to format tables, insert pictures, format text so that you can start using Latex immediately. The main advantage of Latex is that you do not need to care about the formatting of the entire document because Latex will do the job for you! Nice!

Latex can be downloaded free here: <http://www.latex-project.org/>

Usually, the first Chapter of a Dissertation has the following headings (see below). Your dissertation may have different ones, so change these as you wish.

1.1 Aims and Objectives

The main question that this dissertation addresses is In order to address this question, this work focuses on the following issues in the context of :

- Firstly, . . .
- Secondly, . . .

¹It is simple to create a footnote.

1. INTRODUCTION

1.2 Methodology

In order to address this question the following approach was taken.

1.3 Research Contribution

The primary research contribution is:

1.4 Thesis Outline

The remaining chapters of this dissertation are as follows:

Chapter Two is an introductory discussion to . . .

Chapter Three is a historical review of . . .

Chapter Four presents the . . .

Chapter Five describes the . . .

Chapter Six draws conclusions and evaluates the . . . Lastly, it suggests possible future works.

A series of documents have been included in the Appendix section of this dissertation. These are:

- *Appendix A* outlines . . .
- *Appendix B* presents . . .
- *Appendix C* includes . . .

Attached to this dissertation is a CD containing the following items:

- *folder 1*: . . .
- *folder 2*: . . .

2

Chapter 2

Latex will automatically renumber sections! Also it creates list of figures, list of tables and references! Nice job!

This document is formatted according to the University of Limerick specification for the submission of a Master or PhD dissertation. Double-page printing is already in the code! If you want to change things you need to modify the ‘PhDthesisPSnPDF.cls’ file found in Latex/Classes

2.1 Font

Normal font **Bold Font** *italics*. List of **colours** **for** **font** can be found at: <http://en.wikibooks.org/wiki/LaTeX/Colors>

2.1.1 Special characters

Special characters such as & and % need to be treated in a special way. Check the full list of special characters here: http://en.wikibooks.org/wiki/LaTeX/Special_Characters

2.1.1.1 Subsubsection Title

2.1.2 Wiki page

You can learn more about Latex by clicking [here](#).

2. CHAPTER 2

2.2 Another Section

2.2.1 Another Subsection

2.2.1.1 Another Subsubsection Title

3

Chapter 3

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).



Figure 3.1: Title of Figure - Description. [Source: (Digitalart 2013)]

3. CHAPTER 3

3.2 Creating a list

- (2000) item 1.
- (2004) item 2.
- (2010) item 3.
- (2013) item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

*There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.*
(Wordsworth 1799, verses 208-218)

3. CHAPTER 3

4

Math formulae in Latex

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol T denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.5}$$

4. MATH FORMULAE IN LATEX

$$\begin{aligned}
&= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{|A_{ba}| |M_{bm}| \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}} \\
&= \frac{-0.2338, -0.0931, -0.0651}{0.2601} \\
&= (-0.8989, -0.3579, -0.2503)
\end{aligned}$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

5

Chapter 5

5.1 References

Open the file ‘references_myphd’ with BibTex. The file is located in ‘/7_references/’

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book:** (Surname et al. 2000, p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter:** (Surname 2001)
- **Book Contribution:** (Surname 2002)
- **MUSIC CD/DVD:** (Surname 2003)
- **Journal Article:** (Surname et al. 2004)
- **Patent:** (Surname 2005)
- **PhD/Master Dissertation:** (Surname 2006)
- **Conference Paper:** (Surname et al. 2007)
- **Technical Report:** (Surname 2008)

5. CHAPTER 5

- **Unpublished Report:** (Surname 2009)
- **Web Videos**(Surname 2010) (. . . not the name of the person who uploaded the video though)
- **Web Paper:** (Surname and Surname 2011)
- **Web Journal Article:** (Surname 2012)
- **Webpage:** (Surname 2013)
- **Downloaded Images or Sounds:** (Digitalart 2013)

You can also you this kind of referencing if needed in the text:
Surname (2001) said that . . .

6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

6. CONCLUSIONS AND FUTURE DIRECTIONS

Appendix A

Insert a figure



Figure 1: Title of Figure - Description. [Source: (Digitalart 2013)]

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 * Copyright (c) 2013, Giuseppe Torre
 * All rights reserved.
 *
 * Redistribution and use in source and binary forms, with or without
 * modification, are permitted provided that the following conditions are met:
 *     * Redistributions of source code must retain the above copyright
 *       notice, this list of conditions and the following disclaimer.
 *     * Redistributions in binary form must reproduce the above copyright
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 *       derived from this software without specific prior written permission.
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 * (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 * LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 * ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 * (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 * SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 */

#include "ext.h"
#include "ext.mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//*****CALIBRATION STUFF*****//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX.OFFSET 2043 //initial offset in accelerometer X
#define ACCELY.OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ.OFFSET 2264//initial offset in accelerometer Z

#define MagX.OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667 // initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
// ***** END CALIBRATION STUFF *****//

void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
    t_object m_ob; // Must always be the first field; used by Max

    Atom m_args[9]; // we want our inlet to be receiving a list of 10 elements

    long m_value; //inlet

    void *m_R1; //these are all the outlets for the 3 X 3 Matrix
    void *m_R2; // R1 -> first top left cell ..R2 middle cell of the first
    row ...and so on
    void *m_R3; //
    void *m_R4; // End Outlets
} t_triad;

void *triad_new(long value);

void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
s);
void triad_free(t_triad *triad);

void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);

void MatrixByMatrix(double *Result, double *MatrixLeft, double *
MatrixRight);

void Matrix2Quat(double *Quat, double *Matrix);
void Quat2Matrix(double *Matrix, double *Quat);
void inverseQuat(double *InvQuat, double *RegQuat);
void NormQuat(double *YesQuat, double *NotQuat);
void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
void NormVect(double *YesVect, double *NotVect);
double orientationMatrix[9];
double Result[9];
int i;
double InorientationMatrix[9];

double temp[6];
double ref[6];
double vectAx[3];
double vectAy[3];
double vectAz[3];
double vectBx[3];
double vectBy[3];
double vectBz[3];
double MagnCrosProd_A;
double MagnCrosProd_B;
double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
VectBynorm, VectBznorm;
```

```

        double m[9], n[9];
        double quat_e[4];
        double invquat_e[4];
        double mult;
        double quat_new[4];
        double quat_old[4];
        double ecs, accex_ADCnumber, y, accey_ADCnumber, z, accez_ADCnumber, mx,
            magnx_ADCnumber, my, magny_ADCnumber, mz, magnz_ADCnumber;

// SLERP Variables

        double trace, Suca;
        double tol[4], omega, sinom, cosom, scale0, scale1, tez,
            orientationMatrixA[9];

int main(void)
{
    // set up our class: create a class definition
    setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
        sizeof(t_triad), 0L, A_GIMME, 0);
    address((method)triad_list, "list", A_GIMME, 0);
    address((method)triad_assist, "assist", A_CANT, 0);
    finder_addclass("Maths", "triad");
    post(".... I 'm-TRIAD-Object !.... from -AHRS- Library ...", 0);
    return 0;
}

/* ----- triad_new ----- */

void *triad_new(long value)
{
    t_triad *triad;
    triad = (t_triad *)newobject(this_class);           // create the new instance and return
        a pointer to it
    triad->m_R4 = floatout(triad);
    triad->m_R3 = floatout(triad);
    triad->m_R2 = floatout(triad);
    triad->m_R1 = floatout(triad);

    return(triad);
}

etc.

etc.

```

Appendix

Bibliography

- digitalart (2013), ‘Recycle Symbol With Tree Stock Image’, in *www.freedigitalphotos.net* [online], available: http://http://www.freedigitalphotos.net/images/Recycling_g385-Recycle_Symbol_With_Tree_p39521.html [accessed: 1 January 2013] . 5, 12, 15
- Surname, N. (2001), Chapter Title, *Book Title*, Publisher, 1-100. 11, 12
- Surname, N. (2002), Title of Book, in editors, eds., *Title of Contribution*, Publishers, 1-100. 11
- Surname, N. (2003), *CD Title* [CD], Label. 11
- Surname, N. (2005), *Title*, U.S. Patent 0,000,001 available: <http://www.websitename.com> [accessed: 1 January 2013]. 11
- Surname, N. (2006), *Title of Dissertation*, PhD thesis, Centre of Research - University of BLA. 11
- Surname, N. (2008), *Title of Report*, Technical report, Name of Institution. 11
- Surname, N. (2009), *Title of Unpublished Report*, name of Institution, unpublished. 12
- Surname, N. (2010), ‘Title of the Video’, *Collection (if any) at* <http://http://vimeo.com/user3594431> [online], available: <http://vimeo.com/44650248> [accessed: 1 January 2013] . 12
- Surname, N. (2012), ‘Title of Web Journal Article’, in *Name of Journal*, **15**(11) available: <http://www.linktojournalarticle.html> [accessed: 1 January 2013] . 12
- Surname, N. (2013), *Title of Webpage* [online], available: <http://www.linktowebsite.com> [accessed: 1 January 2012]. 12

BIBLIOGRAPHY

Surname, N. and Surname, N. (2011), ‘Title of the article’, *Name of Conference or Organisation publishing*, available: <http://www.article.pdf> [accessed: 1 January 2011] . 12

Surname, N., Surname, N. and Surname, N. (2000), *Title of Book*, Publisher. 11

Surname, N., Surname, N. and Surname, N. (2004), ‘Title of Journal Article’, *Journal name*, **9**(2), 1–100. 11

Surname, N., Surname, N., Surname, N. and Surname, N. (2007), ‘Title of Paper’, in editor name, *In Proceedings of Conference name*, City, Month 22-28, 2007, City, Country, 1-10. 11

Wordsworth, W. (1799), ‘The Prelude’, *Name of Conference or Organisation publishing*, available: <https://genius.com/William-wordsworth-the-prelude-book-12-annotated> [accessed: 1 January 2011] . 7